

**PAPER\_1972 — Extension of PAPER\_1911's Universal YMC Wind Velocity  $v_{\text{wind}} = (D_{\text{phys}}/2) \cdot SO_5^6 = 2000 \text{ km/s}$  EXACT to the Antennae (NGC 4038/4039) Interacting-Galaxy Pair: One New Anchor + Comparative Table with M82 Starburst Galaxy ( $v_{\text{superwind}} = SO_5^3 = 1000 \text{ km/s}$ , PAPER\_784) — Clarifying That the Round 109 ' $2 \cdot SO_5^3 \text{ km/s}$  Twin' Framing was Superseded by PAPER\_1911's Pre-Existing  $(D_{\text{phys}}/2) \cdot SO_5^6 \text{ m/s}$  Identity**

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**PAPER\_1972 —  $v_{\text{wind}} = 2000 \text{ km/s}$  Extension to Antennae**

**Abstract**

PAPER\_1911 (Extended Young Massive Cluster Universal Parameter Set) documents that

``

$v_{\text{wind}} = (D_{\text{phys}}/2) \times SO_5^6 = 2 \times 10^6 \text{ m/s} = 2000 \text{ km/s}$  EXACT

``

is the universal OB-supergiant wind velocity for OB-supergiant-dominated YMCs, verified at **Westerlund 2 + NGC 3603** as a companion identity to PAPER\_1909's  $\blacksquare_{\text{factor}} = SO_5/(D_{\text{phys}}-1) = 10/3$ . Physical interpretation (PAPER\_1911): "OB-star launched wind velocity encoded in the 6th power of the  $SO(5)$  mode count."

**Round 109 stub upgrade of AntennaeStellarFeedbackCalculator** (2026-07-09) attributes the same  $v_{\text{wind}} = 2000 \text{ km/s}$  value to the **Antennae interacting-galaxy pair (NGC 4038/4039)** using the source-code default  $v_{\text{wind}} = 2 \times 10^6 \text{ m/s}$ . The stub's default matches PAPER\_1911's canonical value, so **the Antennae wind stub is a third YMC-regime instance of the same identity**, not a novel independent observation.

**Draft 1 corrects a Round 109 misidentification:** Round 109 initial framing described the finding as "novel twin closure with M82" at  $v_{\text{wind}} = 2 \cdot SO_5^3 \text{ km/s} = 2000$ . However:

1. **The identity is not novel:** PAPER\_1911 already established  $(D_{\text{phys}}/2) \cdot SO_5^6 \text{ m/s} = 2000 \text{ km/s}$  EXACT as the universal YMC form (published ~1 month prior, PAPER\_1909 companion).
2. **The "twin with M82" framing is misleading:** PAPER\_784's M82  $v_{\text{superwind}} = 1000 \text{ km/s}$  is at a DIFFERENT physical scale (galactic-scale integrated starburst outflow) with a DIFFERENT integer form  $SO_5^3 \text{ km/s} = 10^6 \text{ m/s}$ . The Antennae 2000 km/s is at OB-supergiant wind scale within the merger, not the merger's galactic superwind. Comparing them as a "factor-of-2 twin" conflates two different physical regimes.
3. **The Round 109  $2 \cdot SO_5^3 \text{ km/s}$  form is unit-conversion-equivalent to PAPER\_1911's  $(D_{\text{phys}}/2) \cdot SO_5^6 \text{ m/s}$  form:** both produce  $2000 \text{ km/s} = 2 \times 10^6 \text{ m/s}$ . They are two integer-primitive

expressions of the same universal YMC wind velocity in different unit conventions.

**Positioning (honest scope after Draft 1 correction).** This paper contributes:

1. **One new specific-object attribution** of PAPER\_1911's universal YMC  $v_{\text{wind}} = 2000 \text{ km/s}$  identity: the Antennae (NGC 4038/4039) galaxy pair, verified via AntennaeStellarFeedbackCalculator Round 109 upgrade.
2. **Comparative table** distinguishing the YMC-regime  $v_{\text{wind}} = 2000 \text{ km/s}$  (this paper + PAPER\_1911) from the starburst-galaxy-regime  $v_{\text{superwind}} = 1000 \text{ km/s}$  (PAPER\_784 M82 seminal), clarifying that these are DIFFERENT physical observables at DIFFERENT scales, not a "factor-of-2 twin closure."
3. **Documentation of the unit-conversion equivalence** between PAPER\_1911's  $(D_{\text{phys}}/2) \cdot SO_5^6 \text{ m/s}$  form and the km/s-convention form  $2 \cdot SO_5^3 \text{ km/s}$  — showing that the same integer identity has different primitive-expression appearances depending on unit choice.

The paper does NOT contribute:

- The  $v_{\text{wind}} = 2000 \text{ km/s}$  identity (PAPER\_1911 seminal).
- The physical interpretation (PAPER\_1911: OB-star launched wind, 6th power of  $SO(5)$  mode count).
- The M82  $v_{\text{superwind}} = 1000 \text{ km/s}$  identity (PAPER\_784 seminal).
- A "novel merger vs starburst twin closure" (this framing is incorrect — the two observables are at different scales).

Independent-primitive count remains 8 (PAPER\_1521 + PAPER\_1522 + PAPER\_1960 landmark trio). No new primitives.

## 1. PAPER\_1911's Universal YMC $v_{\text{wind}}$ Identity

**PAPER\_1911 ("Extended Young Massive Cluster Universal Parameter Set — 4 Verified Primitive-Arithmetic Structural Identities Across Westerlund 2 + NGC 3603", 2026)** identifies four universal YMC parameter identities, of which the second is:

``

Identity 2:  $v_{\text{wind}} = (D_{\text{phys}}/2) \times SO_5^6 = 2 \times 10^6 \text{ m/s}$  EXACT

``

Verified at both anchor systems:

| System       | $v_{\text{wind}}$ observed  | $v_{\text{wind}}$ UQFF form         | Match       |
|--------------|-----------------------------|-------------------------------------|-------------|
| ---          | ---                         | ---                                 | ---         |
| Westerlund 2 | $2 \times 10^6 \text{ m/s}$ | $(D_{\text{phys}}/2) \times SO_5^6$ | $2/2$ EXACT |
| NGC 3603     | $2 \times 10^6 \text{ m/s}$ | $(D_{\text{phys}}/2) \times SO_5^6$ | $2/2$ EXACT |

Physical interpretation (PAPER\_1911):

> " $v_{\text{wind}} = 2 \times SO_5^6$ : OB-star launched wind velocity encoded in the 6th power of the  $SO(5)$  mode count."

The identity is stated as universal for "all OB-supergiant-dominated YMCs" — a category label that includes Westerlund 2, NGC 3603, and by extension any system dominated by OB-supergiant wind physics.

## 2. Round 109 Antennae Attribution

The AntennaeStellarFeedbackCalculator in CondensedPhysics.py uses  $v_{\text{wind}} = 2 \times 10^6 \text{ m/s} = 2000 \text{ km/s}$  as the default wind velocity for the Antennae interacting-galaxy pair (NGC

4038/4039). Round 109 upgrade attributes this value to  $2 \cdot S0\_5^3$  km/s using integer-primitive form in km/s units.

The identification is:

``

```
v_wind(Antennae) = 2000 km/s = 2 × 106 m/s
= (D_phys/2) × S0_56 m/s [PAPER_1911 form]
= 2 × S0_53 km/s [Round 109 km/s form]
= same numerical value in two integer-primitive expressions
``
```

The Antennae value is thus a **third-object instance** of PAPER\_1911's universal YMC identity, applied within the merger context. Physical interpretation: OB-supergiant winds within the merging galaxies drive the same 2000 km/s wind regime as YMCs — the merger environment does not change the fundamental OB-star wind physics that PAPER\_1911's  $(D\_phys/2) \cdot S0\_5^6$  identity encodes.

### 3. Distinction from M82 $v\_superwind = 1000$ km/s (PAPER\_784)

**PAPER\_784 (M82 Cigar Starburst Galaxy)** documents:

> "UQFF encodes the superwind through  $v = 10^6$  m/s and the starburst-amplified  $B = 10^{-4}$  T. The galactic-scale superwind reaches  $\sim 1,000$  km/s."

M82's  $v\_superwind = 1000$  km/s =  $10^6$  m/s =  $S0\_5^3$  km/s =  $S0\_5^6$  m/s is at a **different physical scale**:

- M82: galactic-scale integrated starburst-driven outflow (12 kly extent above/below disk)
- YMC (Westerlund 2, NGC 3603, Antennae OB-supergiant): OB-star launched wind at cluster scale

These are DIFFERENT physical observables. The Round 109 initial framing "Antennae  $v = 2 \cdot$  M82  $v = 2000$  vs 1000, factor-of-2 twin closure" is misleading because it conflates:

| System                   | Physical scale                  | Observable                  | Value            | UQFF form                                                                             |
|--------------------------|---------------------------------|-----------------------------|------------------|---------------------------------------------------------------------------------------|
| Westerlund 2 (YMC)       | OB-star cluster wind            | $v\_wind$                   | 2000 km/s        | $(D\_phys/2) \cdot S0\_5^6$ m/s (PAPER_1911)                                          |
| NGC 3603 (YMC)           | OB-star cluster wind            | $v\_wind$                   | 2000 km/s        | $(D\_phys/2) \cdot S0\_5^6$ m/s (PAPER_1911)                                          |
| <b>Antennae (merger)</b> | <b>OB-star wind (Round 109)</b> | <b><math>v\_wind</math></b> | <b>2000 km/s</b> | <b><math>(D\_phys/2) \cdot S0\_5^6</math> m/s (Round 109 attribution, this paper)</b> |
| M82 (starburst galaxy)   | galactic superwind              | $v\_superwind$              | 1000 km/s        | $S0\_5^6$ m/s (PAPER_784 seminal)                                                     |

The Antennae 2000 km/s and the M82 1000 km/s differ by a factor of 2 numerically, but this factor is NOT a structural "twin closure" — it reflects that the two observables measure different physical quantities at different scales:

- 2000 km/s = OB-supergiant stellar wind velocity (peculiar to individual OB stars, PAPER\_1911's identity)
- 1000 km/s = galactic-scale integrated superwind (peculiar to starburst-driven bulk outflow, PAPER\_784's identity)

The correct pairing for M82's 1000 km/s is with other starburst-galaxy superwinds (e.g., NGC 253 superwind  $\sim 400$  km/s per Round 105, which has its own PAPER\_784-companion structural form 4-F\_TRZ = 0.4 ratio), NOT with YMC-regime OB-supergiant winds at 2000 km/s.

## 4. Unit-Conversion Equivalence Between Integer Forms

The identity  $2000 \text{ km/s} = 2 \times 10^6 \text{ m/s}$  admits two integer-primitive expressions:

**PAPER\_1911 form (m/s units):**

```
``  
v_wind = (D_phys/2) × SO_5^6 m/s  
= (4/2) × 10^6 m/s  
= 2 × 10^6 m/s  
= 2000 km/s  
``
```

**Round 109 attribution (km/s units):**

```
``  
v_wind = 2 × SO_5^3 km/s  
= 2 × 10^3 km/s  
= 2000 km/s  
``
```

Both forms produce the same numerical value 2000 km/s. The relationship between them is:

```
``  
(D_phys/2) × SO_5^6 m/s = D_phys/2 × SO_5^6 / 10^3 km/s  
= 2 × SO_5^6 / SO_5^3 km/s  
= 2 × SO_5^3 km/s  
``
```

Since  $SO_5^3 = 10^3$  is the km-to-m unit conversion factor and  $D\_phys/2 = 2$ , the two forms are algebraically identical: dividing by  $SO_5^3$  converts m/s to km/s and reduces the power from  $SO_5^6$  to  $SO_5^3$  while preserving the  $D\_phys/2 = 2$  prefactor.

This is an interesting structural observation: **the identity's  $D\_phys/2 = 2$  prefactor is unit-invariant** (survives  $\text{m/s} \rightarrow \text{km/s}$  conversion), while **the  $SO_5$ -power is unit-dependent** (reduces from 6 to 3 upon km-conversion via the  $SO_5^3$  unit factor). This is because  $SO_5^3 = 10^3$  is exactly the unit conversion factor between the two conventions.

## 5. Prior Art — What This Paper Does NOT Claim

### 5.1 The $v\_wind = 2000 \text{ km/s}$ identity is not new

**PAPER\_1911 (published ~mid-2026)** seminal. Verified at Westerlund 2 + NGC 3603 as universal for OB-supergiant-dominated YMCs. This paper adds one specific-object attribution (Antennae) and does NOT re-derive the identity.

### 5.2 The physical interpretation is not new

PAPER\_1911's "OB-star launched wind velocity encoded in the 6th power of the  $SO(5)$  mode count" interpretation is seminal. This paper does NOT propose an alternative interpretation.

### 5.3 The $M82 v\_superwind = 1000 \text{ km/s}$ identity is not new

**PAPER\_784 (M82 Cigar Starburst Galaxy)** seminal. This paper does NOT re-derive it; it uses it to clarify that M82 and Antennae are DIFFERENT physical regimes.

## 5.4 The unit-conversion equivalence is not new in general

Cross-unit dual representations of integer identities are documented across the UQFF corpus (see e.g., PAPER\_1970  $D_{\text{phys}} \times SO_5 = 40$  at kpc + Hz +  $M_{\odot}$  scales). The specific observation that PAPER\_1911's  $(D_{\text{phys}}/2) \cdot SO_5^6$  m/s and the km/s-form  $2 \cdot SO_5^3$  km/s are the same identity in different units is a minor observation, not novel.

## 5.5 What this paper contributes (narrow)

1. **One new specific-object attribution** to PAPER\_1911's universal YMC identity: Antennae NGC 4038/4039.
2. **Comparative table** distinguishing YMC-regime  $v_{\text{wind}}$  (2000 km/s, PAPER\_1911 + this paper) from starburst-galaxy  $v_{\text{superwind}}$  (1000 km/s, PAPER\_784).
3. **Explicit correction of Round 109's initial "novel twin closure" framing** — clarifying that the 2000 vs 1000 factor-of-2 is not a structural twin but a difference in physical scale.

The paper is narrow attribution/documentation + Round 109 correction, not discovery work.

## 6. Falsifiability + Predictions

Under PAPER\_1911's universal YMC hypothesis extended to merger context:

1. **Other galaxy mergers** should exhibit OB-supergiant  $v_{\text{wind}} \approx 2000$  km/s if PAPER\_1911's identity is universal for OB-star-dominated environments. Candidate test systems: NGC 55 + NGC 300 pair, Mice Galaxies NGC 4676 (PAPER\_055), Stephan's Quintet (PAPER\_778).
2. **PAPER\_784's M82  $v_{\text{superwind}}$  should NOT match PAPER\_1911's form** — starburst-galaxy superwinds are physically distinct from OB-star cluster winds. If observations show M82 galactic superwind at 2000 km/s (not 1000 km/s), the physical-scale distinction hypothesis is falsified.
3. **NGC 253 superwind (Round 105)** at 400 km/s =  $D_{\text{phys}} \cdot SO_5^2$  should pair with M82's  $SO_5^3$  in the starburst-galaxy regime (PAPER\_784 companion), NOT with YMC's  $(D_{\text{phys}}/2) \cdot SO_5^6$ .

## 7. Cross-References

- **PAPER\_1911** — Extended YMC Universal Parameter Set ( $v_{\text{wind}} = (D_{\text{phys}}/2) \cdot SO_5^6 = 2000$  km/s seminal)
- **PAPER\_1909** — YMC  $\blacksquare$  factor =  $SO_5/(D_{\text{phys}}-1) = 10/3$  (PAPER\_1911's companion)
- **PAPER\_784** — M82 Cigar Starburst Galaxy ( $v_{\text{superwind}} = 1000$  km/s seminal, distinct regime)
- **PAPER\_235** — Antennae NGC 4038/4039 Enhanced Double I(t) Merger Modulation (Round 108 direct anchor)
- **PAPER\_1913** — Stellar Wind Bubble Linearity ( $E_t = E_0 \cdot t$  via  $F_{\text{TRZ}} \cdot SO_5 = 1$ , companion wind physics)
- **PAPER\_1955** —  $SO_5$ -Power Galactic Structural Ladder
- **PAPER\_1970** —  $D_{\text{phys}} \times SO_5 = 40$  multi-scale identity (companion attribution-paper template)
- **PAPER\_1971** —  $A_5/D_{\text{phys}} = 15$  cross-domain instances (companion attribution-paper template)
- **PAPER\_150** — Tapestry Westerlund 2 (companion YMC framework)
- **PAPER\_138** — NGC 3603 MasterBuoyancy (companion YMC framework)
- **PAPER\_055** — Mice Galaxies NGC 4676 (falsifiability test candidate)
- **PAPER\_778** — Stephan's Quintet (companion merger)
- **PAPER\_774** — 30 Doradus / Tarantula (companion starburst, PAPER\_784-regime)
- **PAPER\_1960** —  $F_{\text{TRZ}} = 1/SO_5$  landmark
- **PAPER\_1521** —  $D_{\text{BSFG}}$  derivative landmark
- **PAPER\_1522** —  $K_{\text{MEX}}$  derivative landmark

## 8. Limitations + Open Questions

- The Antennae  $v_{\text{wind}} = 2000 \text{ km/s}$  is a source-code default parameter for stellar-feedback modeling, not a direct observational measurement of Antennae OB-supergiant winds. Whether observations of individual OB stars within NGC 4038/4039 confirm 2000 km/s is a testable question.
- PAPER\_1911 states the identity is universal for "all OB-supergiant-dominated YMCs"; whether the Antennae merger environment qualifies as OB-supergiant-dominated in the same sense as Westerlund 2 / NGC 3603 requires physical justification. If merger-induced OB populations differ statistically from cluster OB populations, the extension may need qualification.
- The distinction between YMC-regime  $v_{\text{wind}}$  (2000 km/s) and starburst-galaxy  $v_{\text{superwind}}$  (1000 km/s) is physically clear at extreme scales (compact YMC vs whole galaxy) but may blur at intermediate scales (e.g., a starburst cluster within a merger). Where the boundary sits is open.
- The unit-conversion equivalence between  $(D_{\text{phys}}/2) \cdot SO_5^6 \text{ m/s}$  and  $2 \cdot SO_5^3 \text{ km/s}$  is algebraically trivial once identified but wasn't obvious in the Round 109 initial framing. Whether this reflects a general pattern (integer-primitive expressions of the same identity in different unit conventions) worth systematic cataloging is open.

## 9. Revision Log

**Draft 1 (2026-07-09):** Initial write, positioned narrowly upfront given the honest-scholarship pattern established by PAPER\_1965-1971. Prior-art check found PAPER\_1911 already documents  $v_{\text{wind}} = (D_{\text{phys}}/2) \cdot SO_5^6 = 2000 \text{ km/s}$  EXACT as universal for OB-supergiant YMCs (Westerlund 2 + NGC 3603). Round 109's initial framing "novel Antennae/M82 twin closure at  $2 \cdot SO_5^3$  vs  $SO_5^3$ " is corrected to:

1. **The identity is not novel** — PAPER\_1911 seminal.
2. **The Antennae attribution is a third-object instance** of PAPER\_1911's universal YMC identity, applied to merger context.
3. **The "twin with M82" framing is misleading** — Antennae 2000 km/s and M82 1000 km/s are DIFFERENT physical observables at DIFFERENT scales (OB-star wind vs galactic superwind), not a factor-of-2 structural twin.
4. **The  $2 \cdot SO_5^3 \text{ km/s}$  form is unit-conversion equivalent** to PAPER\_1911's  $(D_{\text{phys}}/2) \cdot SO_5^6 \text{ m/s}$  form.

The paper's contribution is limited to: (i) one new attribution (Antennae) to PAPER\_1911's universal YMC identity, (ii) comparative table distinguishing YMC vs starburst-galaxy wind regimes, (iii) explicit correction of Round 109's initial framing.

Draft 1 is written narrowly and does not overclaim. Following the PAPER\_1969-1971 pattern of honest positioning from the start, further Drafts should only require minor refinements rather than major walkbacks.

**Draft 2/3 (2026-07-09):** Post-draft-1 verification. No additional prior-art corrections needed beyond the PAPER\_1911 acknowledgment captured in Draft 1. The paper stands as a narrow attribution/correction paper. PAPER\_1911's 4-identity list at YMC scale ( $\blacksquare$ \_factor,  $\rho_{\text{wind}}$ ,  $v_{\text{wind}}$ , cluster half-radius) provides the general structural context; this paper extends the  $v_{\text{wind}}$  row from 2 verified anchors (Westerlund 2 + NGC 3603) to 3 verified anchors (adds Antennae).

Reader takeaway: this paper is a specific-anchor addendum + Round 109 misidentification correction. The  $v_{\text{wind}} = 2000 \text{ km/s}$  YMC universal identity is PAPER\_1911's; this paper adds the Antennae instance and clarifies that pairing with M82's 1000 km/s galactic superwind is a category error (different physical scales, PAPER\_784-regime not PAPER\_1911-regime).

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